TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

D.C. Circuits: KCL, KVL and the Superposition Theorem

Three rules run every d.c. network: currents balance at a node, voltages balance around a loop, and with several sources each one acts alone and the results simply add.

EEE-2204 · ELECTRICAL & ELECTRONICS SESSIONAL · EXPERIMENT 1 · LEVEL 2 TERM 2

1. Aim and theory

Aim: to verify experimentally (i) Kirchhoff's current law (KCL), (ii) Kirchhoff's voltage law (KVL) and (iii) the superposition theorem in d.c. resistive circuits.

Kirchhoff's current law is a statement of the conservation of charge. At any node (junction) of a circuit, charge cannot accumulate, so the algebraic sum of the currents at the node is zero — the currents entering a node equal the currents leaving it:

$$\Sigma I = 0 \quad \cdot \quad I_{in} = I_{out}$$

Kirchhoff's voltage law is a statement of the conservation of energy. Around any closed loop, the algebraic sum of the voltages is zero: the rises across sources equal the drops across the resistances, which for a single loop of resistors fed by one source reads

$$\Sigma V = 0 \quad \cdot \quad E = I{\cdot}R_1 + I{\cdot}R_2 + I{\cdot}R_3 + \dots$$

Superposition theorem. In a linear network containing several independent sources, the current in any branch is the algebraic sum of the currents produced by each source acting *alone*, with every other voltage source replaced by its internal resistance — for an ideal source, a short circuit (a current source would be open-circuited; none is used here). The theorem works because the circuit is linear: each response is proportional to the source causing it, so responses add. It applies to currents and voltages, but **not** to power, which depends on the square of the current.

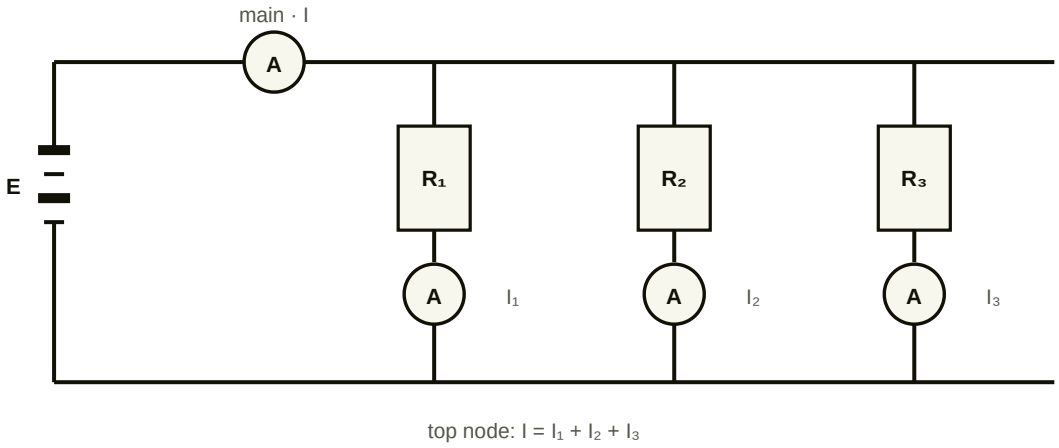


Figure 1 — KCL circuit. One source feeds three parallel branches between two rails; an ammeter in each branch and one in the main line. Whatever the main ammeter reads, the three branch readings must add up to it.

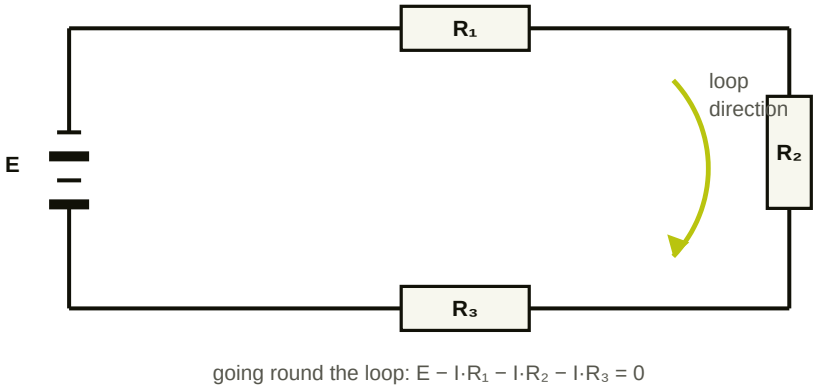
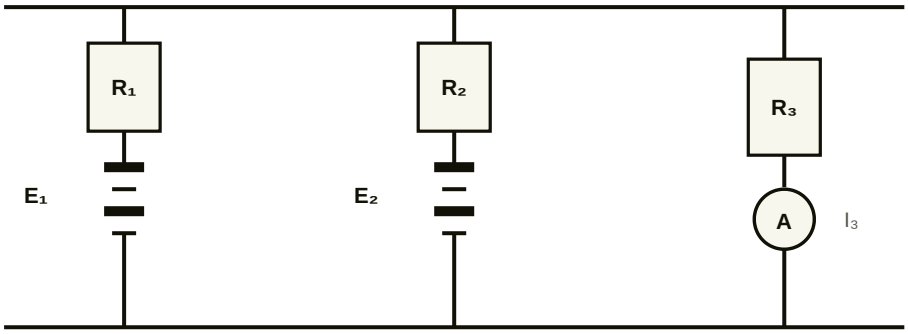


Figure 2 — KVL circuit. A single series loop. A voltmeter moved from resistor to resistor reads drops that must add up to the source voltage, because the same current flows through all three.

2. Apparatus and procedure

- **Apparatus:** regulated d.c. power supply (or battery eliminator), two d.c. voltmeters, four d.c. ammeters (or one ammeter moved between branches), three resistance boxes or fixed resistors of known value, a second d.c. source for the superposition test, connecting wires, SPDT switches.
- **KCL:** wire Figure 1 with R_1 = 6 Ω, R_2 = 3 Ω, R_3 = 2 Ω and set E = 12 V. Read the main ammeter and the three branch ammeters. Change E to 9 V and 6 V and repeat.
- **KVL:** wire Figure 2 with R_1 = 2 Ω, R_2 = 4 Ω, R_3 = 6 Ω in series across 12 V. Read the loop current, then measure the voltage across each resistor in turn.
- **Superposition:** wire the two-source circuit of Figure 3 (E_1 = 12 V with R_1 = 4 Ω, E_2 = 6 V with R_2 = 6 Ω, load R_3 = 3 Ω). Measure the current through R_3 with *both* sources connected. Then disconnect E_2, short its terminals, and measure again (E_1 alone). Restore E_2, disconnect and short E_1,

and measure once more (E_2 alone). Mind the *direction* of each partial current — with one source shorted, the current in R_3 keeps the same downward sense in this circuit, so the parts add.



E_1 branch · E_2 branch · load branch R_3 — all in parallel

Figure 3 — Superposition circuit. Two sources, each with its series resistor, feed a common load R_3. Each source is tested alone with the other short-circuited; the two partial currents in R_3 must add to the current with both sources live.

TEST	READINGS (TYPICAL)	CHECK
KCL, E = 12 V	I = 12.0 A; I_1 = 2.0, I_2 = 4.0, I_3 = 6.0 A	2.0 + 4.0 + 6.0 = 12.0 A ✓
KVL, E = 12 V	I = 1.0 A; V_1 = 2.0, V_2 = 4.0, V_3 = 6.0 V	2.0 + 4.0 + 6.0 = 12.0 V ✓
Superposition	I_3 (both) = 1.78 A; I_3' (E_1 alone) = 1.33 A; I_3'' (E_2 alone) = 0.44 A	1.33 + 0.44 = 1.77 ≈ 1.78 A ✓

3. Worked calculation

Take the superposition circuit of Figure 3: E_1 = 12 V, R_1 = 4 Ω; E_2 = 6 V, R_2 = 6 Ω; R_3 = 3 Ω.

Both sources (direct check by KCL). Let V be the voltage across the parallel group. Each source branch pushes a current (E − V)/R into the top node and the load branch draws V/R_3 out of it. KCL at the node — what flows in equals what flows out — gives V from the node equation:

$$V \times (1/4 + 1/6 + 1/3) = 12/4 + 6/6 \rightarrow V \times 0.75 = 4.0 \rightarrow V = 5.33\ \text{V}$$

$$I_3 = V / R_3 = 5.33 / 3 = 1.78\ \text{A}$$

E_1 alone (E_2 shorted): $R_2 \parallel R_3 = (6 \times 3)/(6 + 3) = 2\ \Omega$, in series with R_1, so the total is 6 Ω and the current from E_1 is 12/6 = 2.0 A. The node voltage is 12 − 2.0 × 4 = 4.0 V, giving $I_3' = 4.0/3 = 1.33$ A.

E_2 alone (E_1 shorted): $R_1 \parallel R_3 = (4 \times 3)/(4 + 3) = 1.71\ \Omega$, total 6 + 1.71 = 7.71 Ω, current from E_2 is 6/7.71 = 0.78 A. Node voltage = 6 − 0.78 × 6 = 1.33 V, giving $I_3'' = 1.33/3 = 0.44$ A.

$$I_3' + I_3'' = 1.33 + 0.44 = 1.77\ \text{A} \approx 1.78\ \text{A (direct)}\ \checkmark$$

The 0.01 A gap is rounding in the intermediate steps; carried at full precision the sum is exact. Note also the branch currents check KCL with both sources live: through R_1, (12 − 5.33)/4 = 1.67 A feeding the node; through R_2, (6 − 5.33)/6 = 0.11 A feeding the node; out through R_3, 1.78 A — and 1.67 + 0.11 = 1.78 A. ✓

4. Precautions

- Observe ammeter and voltmeter polarity; a d.c. meter driven backwards can be damaged, and a reversed reading silently breaks the sign bookkeeping that KCL and superposition depend on.
- Choose meter ranges above the expected value first, then step down — resistor current ratings and meter movements both dislike surprises.
- When a source is "removed" for superposition, replace it with a plain short link, not an open circuit: a real source's internal resistance stays in the circuit, and for the bench supply used here the short reproduces its off-state closely enough.
- Do not test superposition on power: $P \neq P' + P''$ because power varies with I^2, and cross terms appear. The theorem is for currents and voltages only.
- Keep the supply on only while reading; wire-wound resistors heat up and their values drift during long runs.

Practice (viva preparation)

1. State KCL and KVL, and name the conservation law behind each.
2. At a node, 3 A and 5 A enter and 6 A leaves by one branch. What current leaves by the only remaining branch, and in which sense?
3. A 24 V source feeds 4 Ω and 8 Ω in series. Use KVL to find the current and the drop across each resistor.
4. Why is a "removed" voltage source short-circuited, not open-circuited, in the superposition test?
5. Can the superposition theorem be used to find the power dissipated in R_3 of Figure 3? Explain briefly.

Answer key

1. KCL: the algebraic sum of currents at a node is zero (conservation of charge). KVL: the algebraic sum of voltages around a closed loop is zero (conservation of energy).
2. 2 A must leave by the remaining branch: 3 + 5 = 6 + 2.
3. I = 24/(4 + 8) = 2.0 A. Drops: 2.0 × 4 = 8 V and 2.0 × 8 = 16 V; 8 + 16 = 24 V, as KVL requires.

4. Removing a source means removing its e.m.f., not its resistance. An ideal voltage source has zero internal resistance, so with its e.m.f. gone it is a short circuit; open-circuiting it would remove a whole conducting path that exists in the real circuit.
5. No. Power goes as I^2, and $(I' + I'')^2 \neq I'^2 + I''^2$. Find the total current by superposition first, then compute $P = I^2R$ from that total.

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